

SMART HOME IoT

IoT Risk Assessment & Mitigation Plan

Module 5 Group Activity | Polytechnic University of the Philippines

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What is a Smart Home IoT System?

A Smart Home IoT system connects everyday household devices to the internet, enabling remote monitoring, automation, and data collection to improve convenience, security, and energy efficiency.

01

Connected Devices

Smart speakers, cameras, thermostats, door locks, lighting, and appliances

02

Communication Layer

Wi-Fi, Bluetooth, Zigbee, Z-Wave protocols linking devices to hubs

03

Cloud & Apps

Remote control via smartphone apps, voice assistants, and cloud dashboards

Identified Risks in Smart Home IoT Systems

Risk 1

HIGH

Unauthorized Access & Hacking

Weak passwords or unpatched firmware expose smart cameras, locks, and thermostats to remote attackers who can spy on or control the home.

Risk 2

HIGH

Privacy & Data Breach

Smart devices continuously collect behavioral data (sleep patterns, routines, voice commands). A breach can expose sensitive personal information.

Risk 3

MEDIUM

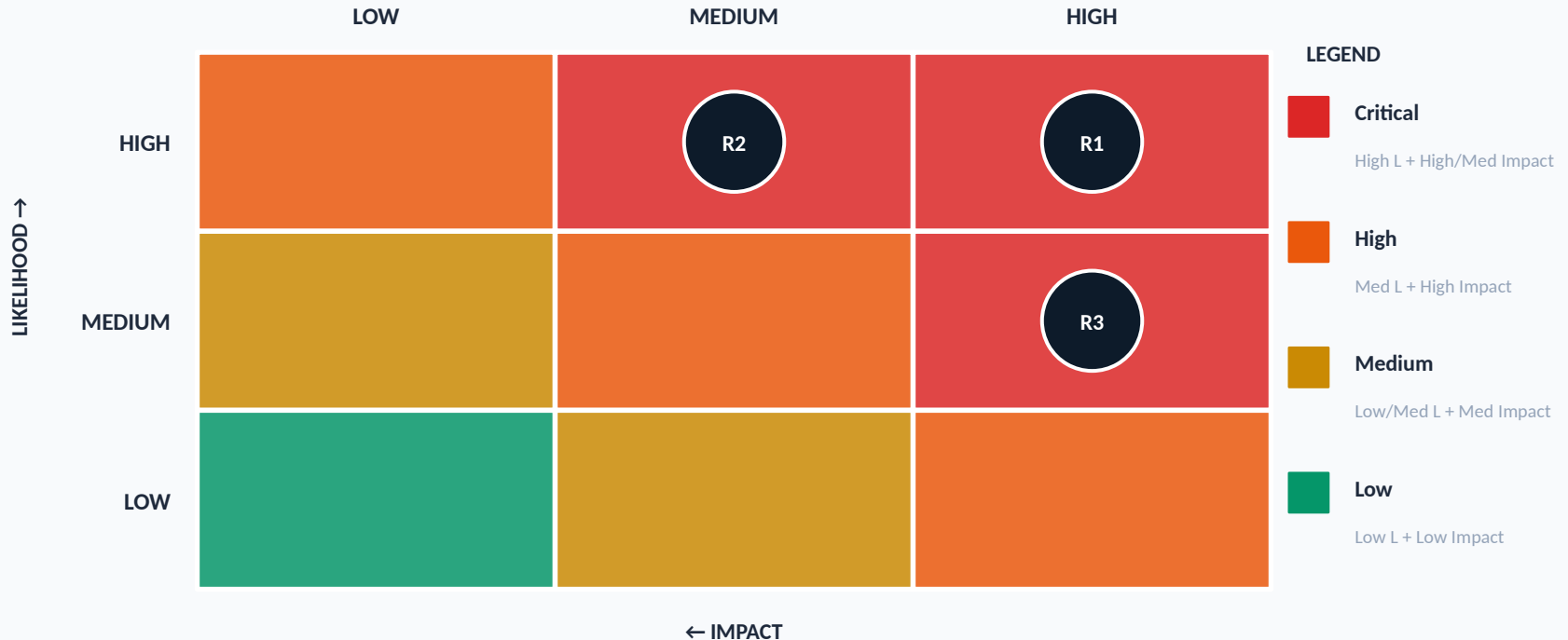
Device Hijacking (Botnets)

Compromised smart home devices can be recruited into botnets and used for large-scale cyberattacks (e.g., DDoS), without the homeowner's knowledge.

RISK MATRIX

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Likelihood vs. Impact Analysis



Proposed Solutions to Identified Risks

01

Risk 01: Unauthorized Access

- ✓ Enable Multi-Factor Authentication (MFA) on all smart home apps and hub accounts
- ✓ Use strong, unique passwords (16+ characters) managed via a password manager
- ✓ Regularly update device firmware and disable unused remote access ports

02

Risk 02: Privacy & Data Breach

- ✓ Encrypt all data in transit (TLS 1.3) and at rest using AES-256 encryption
- ✓ Review and restrict app permissions — devices should only collect necessary data
- ✓ Set up a separate VLAN/guest network for IoT devices to isolate them from personal data

03

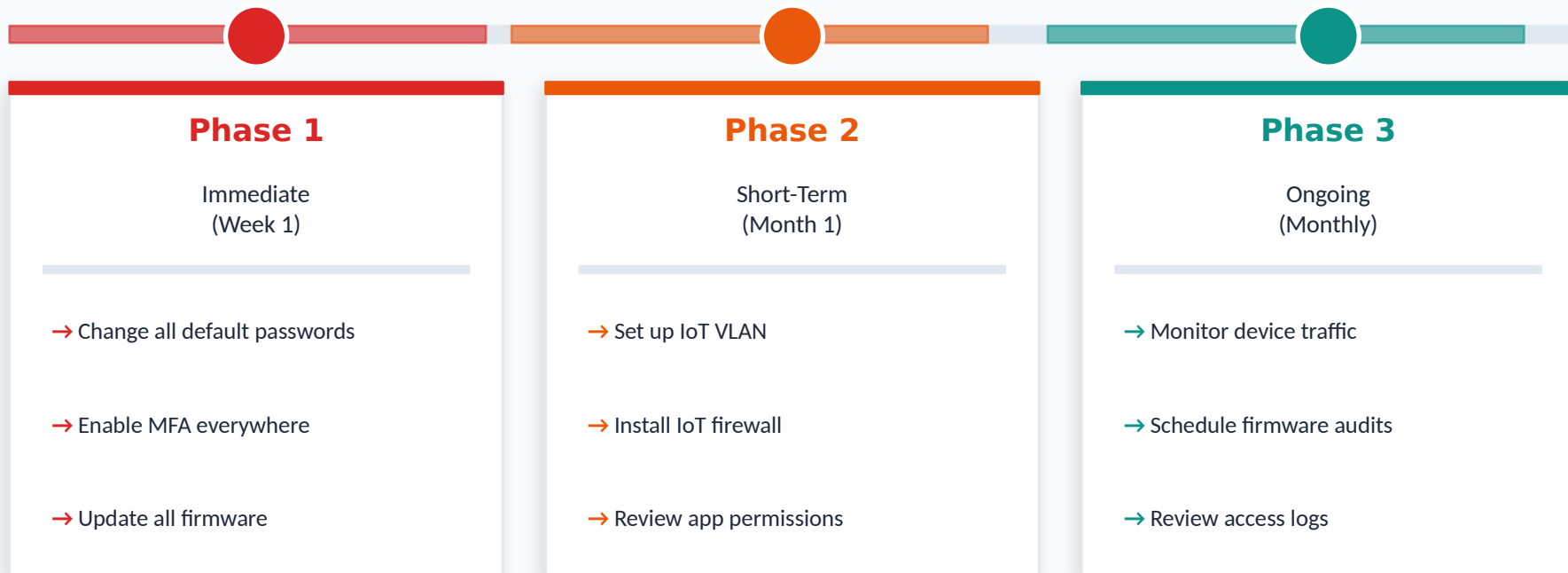
Risk 03: Device Hijacking

- ✓ Deploy a dedicated IoT firewall and intrusion detection system (IDS) on the home network
- ✓ Continuously monitor device traffic for unusual patterns using network monitoring tools
- ✓ Replace default manufacturer credentials immediately upon device setup

IMPLEMENTATION ROADMAP

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Phased Action Plan for Smart Home Security



CONCLUSION

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Smart Home IoT systems offer tremendous convenience but introduce significant security vulnerabilities. By identifying and proactively addressing these risks, homeowners can enjoy a connected lifestyle without compromising their safety and privacy.

3

Critical Risks
Identified

9

Specific
Solutions

3

Implementation
Phases

"Security is not a product, but a process." — IoT best practices begin at home.